

VALLEYVIEW AGDM, AB, Canada

WMO: 712770

Lat: 55.0981N

Lon: 117.1992W

Elev: 2290

StdP: 13.52

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.5	-17.6	-30.0	1.1	-23.2	-23.7	1.6	-17.2	26.4	34.8	23.6	32.8	6.6	170	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	81.9	61.4	78.6	60.1	75.6	58.8	64.4	77.1	62.4	74.6	60.6	72.0	8.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	82.7	68.9	57.7	77.4	66.6	56.0	72.7	64.4	30.7	76.9	29.2	74.4	27.9	72.2	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	19.0	17.0	-31.9	88.1	7.4	3.6	-37.2	90.7	-41.5	92.8	-45.6	94.8	-51.0	97.4		
(5)			-32.1	67.5	7.3	2.3	-37.4	69.2	-41.7	70.6	-45.7	71.9	-51.1	73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	14.6	16.1	25.5	38.6	50.6	57.5	61.9	59.3	51.0	39.7	23.2	13.1		
	DBStd	20.73	15.97	14.93	13.14	9.09	7.92	5.50	5.07	6.00	8.05	8.89	12.67	14.52		
	HDD50	5619	1097	950	761	352	88	6	0	4	81	333	805	1142		
	HDD65	10017	1562	1370	1226	793	448	232	124	192	422	786	1255	1607		
	CDD50	1134	0	0	0	9	106	232	368	294	112	13	0	0		
	CDD65	53	0	0	0	0	1	7	26	16	3	0	0	0		
	CDH74	1012	0	0	0	5	69	157	416	272	90	3	0	0		
	CDH80	197	0	0	0	0	10	21	94	55	17	0	0	0		
(14) Wind	WSAvg	8.6	9.4	8.4	8.8	9.1	8.9	8.2	7.7	7.5	8.5	9.1	8.9	9.3		
(15) Precipitation	PrecAvg	19.1	1.3	0.7	0.7	0.8	1.9	3.3	3.1	2.4	1.6	1.0	1.1	1.0		
	PrecMax	27.1	3.7	1.6	1.4	1.7	3.9	5.9	4.8	5.2	4.6	2.1	2.9	2.0		
	PrecMin	12.6	0.3	0.2	0.2	0.1	0.7	1.7	1.1	0.9	0.2	0.3	0.3	0.2		
	PrecStd	3.6	0.9	0.4	0.4	0.4	0.9	1.1	1.2	1.1	1.0	0.5	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	46.0	53.8	70.9	81.0	82.9	87.8	85.7	82.4	70.6	54.1	43.8		
		MCWB	37.8	37.7	42.1	50.5	56.2	61.2	64.3	62.7	60.0	53.6	42.8	37.4		
	2%	DB	42.0	42.0	48.5	63.0	75.0	78.4	82.2	80.5	76.0	61.6	46.3	39.8		
		MCWB	36.3	35.3	38.8	46.2	54.2	58.8	62.7	61.9	57.8	49.1	39.2	34.8		
	5%	DB	39.1	38.7	44.9	57.0	70.7	74.8	78.5	76.6	70.6	56.3	41.6	36.2		
		MCWB	34.4	33.2	36.8	43.5	51.6	57.2	61.4	59.8	55.4	45.5	36.3	31.8		
	10%	DB	35.7	35.0	41.4	52.6	66.5	71.0	75.1	73.1	64.9	52.3	38.1	32.6		
		MCWB	32.3	30.8	34.7	41.1	49.8	55.6	60.3	58.9	52.5	43.2	33.8	29.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	38.2	43.0	51.2	58.1	63.6	67.8	66.1	61.8	54.1	43.6	37.9		
		MCDB	44.3	44.3	52.6	69.3	75.9	77.2	80.3	79.5	77.8	68.2	52.0	42.8		
	2%	WB	36.7	35.6	39.4	47.5	55.1	61.0	65.3	63.8	59.0	49.7	39.7	35.0		
		MCDB	41.2	41.2	47.1	60.8	71.0	74.2	77.8	76.6	73.3	59.6	45.3	39.2		
	5%	WB	34.9	33.6	37.2	44.1	53.0	59.0	63.0	61.8	56.4	46.7	36.8	32.1		
		MCDB	38.9	38.0	44.4	55.6	67.5	71.0	75.0	73.5	68.2	54.4	40.9	35.6		
	10%	WB	32.0	31.0	35.0	41.7	51.0	57.1	61.3	59.9	53.9	44.1	34.1	29.3		
		MCDB	35.5	35.0	41.0	51.4	64.2	68.5	72.4	70.2	63.4	50.8	37.6	32.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	17.9	19.3	19.6	22.8	21.2	22.1	22.4	21.4	17.5	13.8	14.1		
		MCDBR	16.0	19.6	21.5	27.3	30.4	28.5	28.7	28.8	30.6	24.4	16.5	17.1		
		MCWBR	12.1	14.0	13.5	14.7	13.5	12.2	12.2	12.2	13.8	13.7	11.3	13.5		
	5% WB	MCDBR	15.2	18.7	20.6	25.9	27.2	24.2	25.6	26.1	28.5	23.3	15.8	17.3		
		MCWBR	11.7	13.6	13.2	14.5	12.8	11.5	12.2	12.3	13.2	13.8	11.6	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.249	0.286	0.330	0.342	0.348	0.349	0.348	0.310	0.283	0.251	0.236		
	taud		2.364	2.433	2.417	2.390	2.409	2.410	2.436	2.422	2.540	2.584	2.461	2.283		
	Ebn at Noon		233	275	285	284	286	285	282	275	274	259	229	203		
	Edh at Noon		16	21	27	33	34	35	33	31	24	18	14	13		
(44) All-Sky Solar Radiation	RadAvg		251	566	1038	1468	1740	1873	1848	1530	1047	573	267	180		
	RadStd		21	45	86	98	131	150	128	121	94	64	21	16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page